

5. Discrete distributions

Let $(\Omega, \mathcal{F}, P)$ be our probability space.

$X: \Omega \mapsto \mathbb{R}$ is a discrete random variable if the possible values of X are countable.

We've already visited

$$X \sim \text{Ber}(p) : P(X=1) = p = 1 - P(X=0)$$

$$X \sim \text{Bin}(n, p) : 0 \leq k \leq n, P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$$

I. Poisson distribution

def. We say that a random variable X follows Poisson distribution with parameter λ if

$$k \in \{0, 1, 2, \dots\} : P(X=k) = e^{-\lambda} \frac{\lambda^k}{k!} \quad \leftarrow \text{We've seen earlier why do we need the } e^{-\lambda} \text{ multiplier.}$$

Notation: $X \sim \text{Poi}(\lambda)$

Prop. $E[X] = \lambda$ & $D^2[X] = \lambda$

proof:

$$\underline{E[X]} = \sum_{k=0}^{\infty} k P(X=k) = \sum_{k=1}^{\infty} k P(X=k) = \sum_{k=1}^{\infty} k e^{-\lambda} \frac{\lambda^k}{k!} =$$

first term is zero

$$= \lambda \sum_{k=1}^{\infty} e^{-\lambda} \frac{\lambda^{k-1}}{(k-1)!} = \lambda \sum_{k=0}^{\infty} e^{-\lambda} \frac{\lambda^k}{k!} = \lambda \underbrace{\sum_{k=0}^{\infty} e^{-\lambda} \frac{\lambda^k}{k!}}_{=1} = \lambda$$

$$\underline{E[X(X-1)]} = \sum_{k=0}^{\infty} k(k-1) e^{-\lambda} \frac{\lambda^k}{k!} = \sum_{k=2}^{\infty} k(k-1) e^{-\lambda} \frac{\lambda^k}{k(k-1)(k-2)!} = \lambda^2 \sum_{k=2}^{\infty} e^{-\lambda} \frac{\lambda^{k-2}}{(k-2)!} = \lambda^2 \underbrace{\sum_{k=0}^{\infty} e^{-\lambda} \frac{\lambda^k}{k!}}_{=1} = \lambda^2$$

first two terms are zeros like before

$$\Rightarrow \underline{D^2[X]} = E[X^2] - E^2[X] = E[X(X-1)] + E[X] - E^2[X] = \lambda^2 + \lambda - \lambda^2 = \lambda \quad \square$$

Thm. (Poisson approximation)

Let $\lambda > 0$ and let p_n be a sequence such that

$$n \geq 1 : 0 < p_n < 1 \text{ and } \lim_{n \rightarrow \infty} n p_n = \lambda.$$

Let Y_n be a sequence of discrete random variables such that $Y_n \sim \text{Bin}(n, p_n)$.

Then

$$\lim_{n \rightarrow \infty} P(Y_n = k) = e^{-\lambda} \frac{\lambda^k}{k!} \text{ for } k \in \{0, 1, 2, \dots\}$$

proof: We will use the following results from Calculus:

- $\lim_{n \rightarrow \infty} np_n = \lambda \Rightarrow \lim_{n \rightarrow \infty} p_n = 0$ (1)

- Let $\{a_n\}_{n \geq 1}, \{b_n\}_{n \geq 1}$ be series such that $a_n \xrightarrow[n \rightarrow \infty]{} 0$, $b_n \xrightarrow[n \rightarrow \infty]{} \infty$ and $a_n b_n \xrightarrow[n \rightarrow \infty]{} x$.
Then $\lim_{n \rightarrow \infty} (1 + a_n)^{b_n} = e^x$. (2)

$$P(Y_n = k) = \binom{n}{k} p_n^k (1-p_n)^{n-k} =$$

$$= \frac{1}{k!} \underbrace{np_n}_{\xrightarrow[n \rightarrow \infty]{\lambda}} \cdot \underbrace{(n-1)p_n}_{\xrightarrow[n \rightarrow \infty]{\lambda}} \cdot \dots \cdot \underbrace{(n-k+1)p_n}_{\xrightarrow[n \rightarrow \infty]{\lambda}} \cdot \underbrace{(1-p_n)^n}_{\xrightarrow[n \rightarrow \infty]{e^{-\lambda}}} \cdot \underbrace{(1-p_n)^{n-k}}_{\xrightarrow[n \rightarrow \infty]{1}} \xrightarrow[n \rightarrow \infty]{\frac{\lambda^k}{k!} e^{-\lambda}}$$

k many terms

we applied (2)
we applied (1)

Example: There are 10 light bulbs in a box, each of them is faulty with probability 0,1 independently. What is the probability that there is at most 1 faulty bulb in the box?

X = number of faulty bulbs

with binomial distribution:

$$P(X \leq 1) = P(X=0) + P(X=1) = \binom{10}{0} 0,1^0 \cdot 0,9^{10} + \binom{10}{1} 0,1^1 \cdot 0,9^9 \approx \underline{\underline{0,7361}}$$

with poisson distribution:

$$P(X \leq 1) = P(X=0) + P(X=1) = e^{-\lambda} \frac{\lambda^0}{0!} + e^{-\lambda} \frac{\lambda^1}{1!} = e^{-1} + e^{-1} \approx \underline{\underline{0,7358}}$$

$\lambda = np = 0,1 \cdot 10 = 1$

II. / Poisson process /

def: We say that $g: \mathbb{R} \rightarrow \mathbb{R}$ is order h^α ($\alpha \geq 0$) if $\lim_{h \rightarrow 0} \frac{g(h)}{h^\alpha} = 0$

Notation: $g = o(h^\alpha)$

Example: $h^2 = o(h)$, since $\lim_{h \rightarrow 0} \frac{h^2}{h} = \lim_{h \rightarrow 0} h = 0$

On the other hand $h \neq o(h)$, $\nexists h \neq o(h)$.

def: Let $\lambda > 0$ be a fixed parameter, and consider random events happening as time pass by (like different random points on the line), satisfying the following conditions:

- $P(\text{in an interval of length } h \text{ exactly 1 event happens}) = \lambda h + o(h)$,
- $P(\text{in an interval of length } h \text{ at least 2 events happen}) = o(h)$,

c) The number of events in disjoint intervals are independent.

Then we say that the events follow a Poisson process / Poi(λ) process.
We call λ the rate of the process.

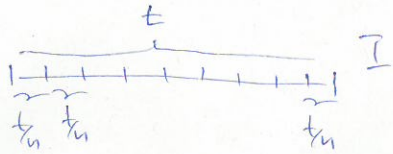
Thm: Consider a Poisson process, and let $N(t)$ be the number of events happening in an interval of length t . Then $N(t) \sim \text{Poi}(t \cdot \lambda)$.

That is

$$k \in \{0, 1, 2, \dots\}, \quad \mathbb{P}(N(t) = k) = e^{-\lambda t} \frac{(\lambda t)^k}{k!}$$

proof: Let I be an interval of length t .

Divide I into n -many intervals of length t_n .



$E_i := \{\text{At least 2 events happen in the } i\text{-th interval}\}$

$F_i := \{\text{Exactly 1 event happens in the } i\text{-th interval}\}$

$$\mathbb{P}(N(t) = k) = \underbrace{\mathbb{P}(\{N(t) = k\} \cap \bigcup_{i=1}^n E_i)}_{\text{I}} + \underbrace{\mathbb{P}(\{N(t) = k\} \cap (\bigcup_{i=1}^n E_i)^c)}_{\text{II}}$$

Now we calculate the probability of these two terms separately.

$$\text{I} \quad \underbrace{\mathbb{P}(\{N(t) = k\} \cap \bigcup_{i=1}^n E_i)}_{0 \leq} \leq \underbrace{\mathbb{P}(\bigcup_{i=1}^n E_i)}_{\substack{\text{inclusion} \\ \uparrow}} \leq \underbrace{\sum_{i=1}^n \mathbb{P}(E_i)}_{\substack{\text{union bound} \\ \uparrow}} = n \cdot \underbrace{\sigma(t_n)}_{\substack{\text{def/b)}}} \xrightarrow[n \rightarrow \infty]{} 0$$

$$\text{II} \quad \mathbb{P}(\{N(t) = k\} \cap (\bigcup_{i=1}^n E_i)^c) = \underbrace{\mathbb{P}(N(t) = k \mid (\bigcup_{i=1}^n E_i)^c)}_{\text{I}} \cdot \underbrace{\mathbb{P}((\bigcup_{i=1}^n E_i)^c)}_{\substack{n \rightarrow \infty \\ \downarrow \\ 1 \text{ by I}}} = \textcircled{*}$$

$$p_n = \mathbb{P}(F_j \mid (\bigcup_{i=1}^n E_i)^c) \stackrel{\text{def/c}}{=} \mathbb{P}(F_j \mid E_j^c) = \frac{\mathbb{P}(F_j)}{\mathbb{P}(E_j^c)} \stackrel{\text{def/a}}{=} \frac{\lambda t_n + \sigma(t_n)}{1 - \sigma(t_n)}$$

By the definition of σ we have

$$n p_n = \frac{\lambda t + n \sigma(t_n)}{1 - \sigma(t_n)} \xrightarrow[n \rightarrow \infty]{} \lambda t$$

$$\textcircled{*} = \binom{n}{k} p_n^k (1 - p_n)^{n-k} \cdot \mathbb{P}((\bigcup_{i=1}^n E_i)^c) \xrightarrow[n \rightarrow \infty]{} e^{-\lambda t} \frac{(\lambda t)^k}{k!} \quad \text{Poi approx.} \quad \square$$

The Poisson process can be generalized to higher dimensions by replacing length with area/volume. If we have a k -dimensional Poisson process with rate λ in a region K with volume $\text{vol}_k(K)$, then

$$\mathbb{P}(N(K) = n) = e^{-\text{vol}_k(K) \lambda} \frac{(\text{vol}_k(K) \cdot \lambda)^n}{n!}$$

and $K_1 \cap K_2 = \emptyset \Rightarrow N(K_1)$ and $N(K_2)$ are independent.

Example 1: I make only 3 grammatical errors in a 60 minutes long speech. What is the probability that I won't do any mistakes in a 30 min lecture? What is the expected number of my errors?

Let's model these grammatical errors as a Poisson process.

$N(t)$ = number of mistakes made in t hours

In 60 mins = 1 hour I make 3 mistakes (on average)

$\Rightarrow \lambda = 3$ is the expected value of $N(1)$.

$$\Rightarrow N(1.5) \sim \text{Poi}(\underbrace{1.5 \cdot 3}_{=4.5}) \Rightarrow P(N(1.5) = 0) = \underline{\underline{e^{-9/2}}}$$

$$E[N(1.5)] = \underline{\underline{9/2}}$$

III. | Geometric distribution

Suppose that we are making independent trials, each having probability $0 < p < 1$ of success. We stop when the first success occurs. Let X be the number of trials required.

def: We say that X follows (optimistic) Geometric distribution with parameter p .

Notation: $X \sim \text{Geo}(p)$, $X \sim \text{Geo}_{\text{opt}}(p)$

$$P(X=n) = \underbrace{(1-p)^{n-1}}_{\substack{\uparrow \\ \text{first } (n-1)\text{-many trials failed}}} p \leftarrow \text{the } n^{\text{th}} \text{ one was the first success}$$

first $(n-1)$ -many trials failed

If Y denotes the number of failures until the first success, then

$$P(Y=n) = P(X=n+1) = (1-p)^n p, \quad n \geq 0 \text{ integer}$$

We say that Y follows pessimistic geometric distribution with parameter p .

Prop: $P(X \geq k) = (1-p)^{k-1}$

proof:
$$P(X \geq k) = \sum_{n=k}^{\infty} P(X=n) = \sum_{n=k}^{\infty} (1-p)^{n-1} p = \sum_{n=0}^{\infty} \underbrace{(1-p)^{n+k-1}}_{\substack{\uparrow \\ \text{geometric sum}}} p = \frac{(1-p)^{k-1}}{1-(1-p)} p = (1-p)^{k-1} \quad \square$$

Prop (Memoryless or "Forever young")

$$P(X \geq n+k | X > n) = P(X \geq k) \text{ for any } k \geq 1$$

proof:

$$\begin{aligned} P(X \geq n+k | X > n) &= \frac{P(\{X \geq n+k\} \cap \{X > n\})}{P(X > n)} = \frac{P(X \geq n+k)}{P(X \geq n+1)} \quad \leftarrow \text{previous prop.} \\ &= \frac{(1-p)^{n+k-1}}{(1-p)^n} = (1-p)^{k-1} \quad \leftarrow \text{previous prop.} = P(X \geq k) \quad \square \end{aligned}$$

Prop: $E[X] = \frac{1}{p}$ and $D^2[X] = \frac{1-p}{p^2}$

proof:

$$E[X] = \sum_{n=1}^{\infty} n P(X=n) = \sum_{n=1}^{\infty} n (1-p)^{n-1} p = p \frac{1}{(1-(1-p))^2} = \frac{1}{p}$$

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \Rightarrow \frac{\partial}{\partial x} \sum_{n=0}^{\infty} x^n = \frac{\partial}{\partial x} \frac{1}{1-x} \Rightarrow \sum_{n=0}^{\infty} n x^{n-1} = \frac{\partial}{\partial x} \frac{1}{1-x} = \frac{1}{(1-x)^2}$$

uniform convergence

$$E[X(X-1)] = \sum_{n=1}^{\infty} n(n-1) (1-p)^{n-1} p = (1-p)p \frac{2}{p^3} = \frac{2(1-p)}{p^2}$$

$$\sum_{n=1}^{\infty} n x^{n-1} = \frac{1}{(1-x)^2} \Rightarrow \sum_{n=1}^{\infty} n(n-1) x^{n-2} = \frac{\partial}{\partial x} \sum_{n=1}^{\infty} n x^{n-1} = \frac{\partial}{\partial x} \frac{1}{(1-x)^2} = \frac{2}{(1-x)^3}$$

uniform convergence

$$\Rightarrow E[X^2] = E[X(X-1)] + E[X] = \frac{2(1-p)}{p^2} + \frac{1}{p} = \frac{2+p}{p^2}$$

$$\Rightarrow D^2[X] = E[X^2] - E[X]^2 = \frac{2+p}{p^2} - \frac{1}{p^2} = \frac{1-p}{p^2} \quad \square$$

IV. | Negative binomial |

Suppose that we keep performing independent trials until the occurrence of the r th success. Each of these trials succeeds with probability $0 < p < 1$. Let X be the number of trials we perform.

def: We say that X follows Negative binomial distribution with parameters p and r .

Notation: $X \sim \text{Neg}(p, r)$

$$P(X=n) = \sum_{1 \leq i_1 < \dots < i_{r-1} \leq n-1} P(i_1, \dots, i_{r-1} \text{ were the successful experiments}) = \binom{n-1}{r-1} p^r (1-p)^{n-r}$$

Prop: Let $X \sim \text{Neg}(p, r)$, then $E[X] = \frac{r}{p}$ and $D^2[X] = r \frac{1-p}{p^2}$.

V. | Hypergeometric distribution |

A sample of size n is chosen randomly from a population of N many objects. In the population, m -many elements have property A , the others don't. Let X be the number of elements with property A in the sample.

def: We say that X follows a Hypergeometric distribution with parameters N, m, n .

Notation: $X \sim \text{Hyp}(N, m, n)$

$$i \in \{0, \dots, m\}; \quad P(X=i) = \frac{\binom{m}{i} \binom{N-m}{n-i}}{\binom{N}{n}}$$

we choose i many from the ones with property A
 we choose the remaining $n-i$ elements from the others
 all possible ways of choosing n elements out of N many.

Prop. Let $X \sim \text{Hyp}(N, m, n)$, then $E[X] = \frac{m \cdot n}{N}$

proof:

$$E[X] = \sum_{i=0}^m i \cdot \frac{\binom{m}{i} \binom{N-m}{n-i}}{\binom{N}{n}} = \sum_{i=1}^m \frac{i \cdot \frac{m!}{i!(m-i)!} \cdot \frac{(N-m)!}{(n-i)!(N-m-n+i)!}}{\frac{N!}{n!(N-n)!}} =$$

$$= \frac{m \cdot n}{N} \sum_{i=1}^m \frac{\binom{m-1}{i-1} \binom{N-m}{n-i}}{\binom{N-1}{n-1}} = \frac{m \cdot n}{N} \sum_{i=0}^{m-1} \underbrace{P(Y=i)}_{=1} = \frac{m \cdot n}{N}$$

$= P(Y=i-1) \text{ for } Y \sim \text{Hyp}(N-1, m-1, n+1)$ □

VI. | Continuous distributions

Let $(\Omega, \mathcal{F}, P)$ be a probability space.

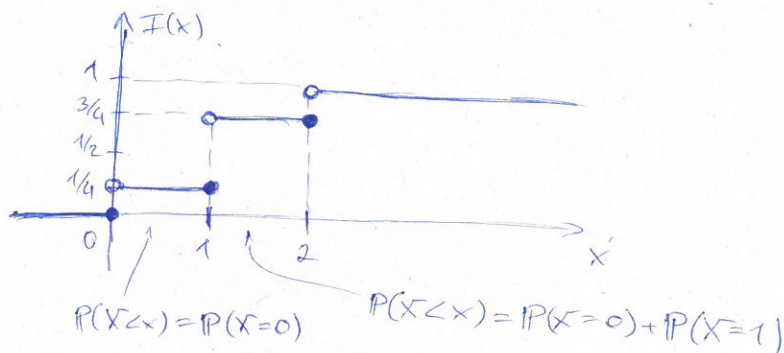
def. Let X be a random variable. We call the function $F(x) = P(X \leq x)$ the distribution function of X .

Example: Toss a coin two times. X = number of heads
What is the distribution function of X ?

$$P(X=0) = 1/4$$

$$P(X=1) = 1/2$$

$$P(X=2) = 1/4$$



Prop. Let X be a random variable. Let $F(x)$ be its distribution function. Then the following hold.

a) F is monotonically increasing

b) $\lim_{x \rightarrow -\infty} F(x) = 0$

c) $\lim_{x \rightarrow \infty} F(x) = 1$

d) F is left continuous

proof:

a) $\forall x < y, \{X < x\} \subseteq \{X < y\} \Rightarrow \forall x < y, P(X < x) < P(X < y)$

b) $\lim_{x \rightarrow -\infty} F(x) = \lim_{x \rightarrow -\infty} P(X < x) = P(\bigcap_{n \in \mathbb{N}} \{X < n\}) = P(\emptyset) = 0$

c) $\lim_{x \rightarrow \infty} F(x) = \lim_{x \rightarrow \infty} P(X < x) = P(\bigcup_{n \in \mathbb{N}} \{X < n\}) = P(\Omega) = 1$

d) $\lim_{x \rightarrow x_0} F(x) = \lim_{n \rightarrow \infty} F(x_0 - \frac{1}{n}) = \lim_{n \rightarrow \infty} P(X < x_0 - \frac{1}{n}) = P(\bigcap_{n=1}^{\infty} \{X < x_0 - \frac{1}{n}\}) = P(\{X < x_0\}) = F(x_0)$ □